

DHCP Lease Renewal

After disabling Dnsmasq and ensuring Kea DHCP was the only active DHCP service, the Ubuntu client lease was renewed.

Commands:

```
```bash id="n5i0jg" sudo dhclient -r ens18
```

```
sudo dhclient ens18
```

Purpose:

```
```text id="xyyl0w"
Release existing DHCP lease

Request a new DHCP lease
```

Verification of DHCP Options

Command:

```
```bash id="j9z5r7" resolvectl dns
```

Output:

```
```text id="gt7i0u"
Link 2 (ens18):

10.1.10.10
```

PASS

Client now receives the correct DNS server.

Verification of Search Domain

Command:

```
```bash id="l3frye" resolvectl domain
```

Output:

```
```text id="mjys6y"  
Link 2 (ens18):  
  
worldskills.ph
```

PASS

Client now receives the correct search suffix.

Hostname Resolution Testing

One of the primary goals of DHCP integration is allowing users to resolve hosts without typing fully qualified domain names.

Instead of:

```
```text id="0w4ptp" int-srv01.worldskills.ph
```

users can simply type:

```
```text id="nmn36v"  
int-srv01
```

Ping Test

Command:

```
```bash id="11mfrd" ping int-srv01
```

Output:

```
```text id="6yrfzh"  
PING int-srv01.worldskills.ph  
(10.1.10.10)
```

PASS

The search suffix automatically expanded:

```
```text id="w8d45y" int-srv01
```

↓

```
```text id="6h6i56"
int-srv01.worldskills.ph
```

Internal DNS Resolution

Command:

```
```bash id="s31f9j" host int-srv01.worldskills.ph
```

Output:

```
```text id="j1w3h6"
int-srv01.worldskills.ph
has address 10.1.10.10
```

PASS

Forward DNS works.

Reverse DNS Resolution

Command:

```
```bash id="a4u7rn" host 10.1.10.10
```

Output:

```
```text id="0vg5up"
10.1.10.10.in-addr.arpa
domain name pointer
```

```
int-srv01.worldskills.ph
```

PASS

Reverse DNS works.

Gateway Reverse Lookup

Command:

```
```bash id="9osgrk" nslookup 10.1.10.1
```

Output:

```
```text id="7mbh1u"
10.1.10.1.in-addr.arpa

name = _gateway
```

PASS

OPNsense successfully registered its gateway record.

Internet DNS Resolution

The next step was validating recursive DNS functionality.

The DNS server must resolve:

```
```text id="3b09w3" Internal domains
```

and

```
```text id="5tqucv"
Public internet domains
```

Test Public DNS Resolution

Command:

```
```bash id="x6xtjp" nslookup google.com
```

Output:

```
```text id="ef37dv"  
google.com
```

```
172.x.x.x
```

```
2404:6800:xxxx
```

PASS

Internet DNS resolution works.

Recursive Resolver Validation

Purpose:

Determine whether the DNS infrastructure can resolve external internet domains.

Command:

```
```bash id="obgfrn" dig @10.1.10.10 google.com
```

Expected:

```
```text id="2tr8t4"  
NOERROR
```

Result:

```
```text id="ec08pd" Successful
```

PASS

The DNS server supports recursive lookups.

---

# DNS Flow Validation

Final architecture:

```text id="clv5v7"

Ubuntu Client

10.1.10.175

|

|

| DNS Query

v

Debian DNS Server

10.1.10.10

(BIND9)

|

+-----+

|

|

v

worldskills.ph

|

|

v

Internet

Services Verified

DHCP

Verified:

```text id="kudys3" Automatic IP assignment

PASS

---

Verified:

```text id="d5b8nb"

Automatic gateway assignment

PASS

Verified:

```text id="s0u37e" Automatic DNS assignment

PASS

---

Verified:

```text id="eh06iv"  
Automatic search suffix assignment

PASS

DNS

Verified:

```text id="ey0mqg" Forward lookup

PASS

---

Verified:

```text id="m12rwu"  
Reverse lookup

PASS

Verified:

```text id="4nhjdc" Internal DNS records

PASS

---

Verified:

```text id="2yuwju"  
External DNS records

PASS

Troubleshooting Summary

Issue #1

```text id="yv22hu" Reverse lookup failed

Cause:

```text id="o7s9zw"  
Incorrect PTR configuration

Resolution:

```text id="1v2dr9" Correct reverse zone Validate Reload

---

Issue #2

```text id="vdcwjg"  
Client DNS resolution failed

Cause:

```text id="z77s3s" Client used 10.1.10.1 instead of 10.1.10.10

Resolution:

```
```text id="4n0t8l"
Investigate DHCP lease
Analyze DNS options
```

Issue #3

```text id="rduud4" Wrong DHCP DNS settings

Cause:

```
```text id="3e2hnj"
Dnsmasq conflict
```

Resolution:

```text id="md5knh" Disable Dnsmasq

Use Kea DHCP only

```

Engineering Lessons Learned

Lesson 1

Always validate DNS zones before restarting services.

Commands:

```bash id="mspq59"
named-checkconf

named-checkzone
```

prevent downtime caused by syntax errors.

Lesson 2

DHCP lease inspection is one of the fastest troubleshooting methods.

Command:

```
```bash id="hzwbg" grep domain-name-servers
```

```
revealed the incorrect DNS server immediately.

Lesson 3

Multiple DHCP services create unpredictable behavior.

Only one service should distribute DHCP information on a subnet.

Lesson 4

Successful DNS testing requires:

```text id="igwudq"
Forward lookup

Reverse lookup

Internal domains

External domains
```

Lesson 5

Enterprise DNS servers should use static IP addresses.

DHCP-assigned DNS servers create instability.

Skills Practiced

Linux Administration

- Service management
- Package installation
- Network configuration

DNS Administration

- BIND9 deployment
- Zone creation
- Forward lookup
- Reverse lookup
- DNS validation

DHCP Administration

- Scope creation
- DNS option delivery
- Search suffix delivery

OPNsense Administration

- Interface configuration
- DHCP configuration
- DNS integration
- Service troubleshooting

Network Troubleshooting

- NXDOMAIN analysis
- SERVFAIL analysis
- DHCP lease analysis
- DNS path validation
- Service conflict investigation

Common Interview Questions

What is DNS?

DNS converts:

```text id="h2l9u7" Hostname

into:

```
```text id="h1w4d5"  
IP Address
```

What is a PTR Record?

PTR records perform:

```text id="6e5yt2" IP Address → Hostname

```
resolution.
```

```

```

## Difference Between Forward and Reverse DNS?

Forward:

```
```text id="m7t42r"  
Hostname → IP
```

Reverse:

```text id="zg64ri" IP → Hostname

```

```

## Why Use Dedicated DNS Servers?

Benefits:

- Centralized management
- Better scalability
- Easier troubleshooting
- Enterprise best practice

```

```

## What Caused The DNS Issue?

Root cause:

```
```text id="qot4hk"
Dnsmasq DHCP
and
Kea DHCP

both distributing DHCP options
```

Result:

```
```text id="qj2h3m" Incorrect DNS server assignment
```

```

Final Validation Checklist

Infrastructure

```text id="syt5g8"
OPNsense
```

PASS

```
```text id="o2f6a4" Debian DNS Server
```

```
PASS

```text id="2o71kh"
Ubuntu Client
```

PASS

DNS

```
```text id="uxy1nb" Forward Zone
```

```
PASS
```

---

```text id="fw02s4"  
Reverse Zone

PASS

```text id="b61ggv" Internal Resolution

PASS

---

```text id="1f91jh"  
External Resolution

PASS

DHCP

```text id="i4spvj" IP Assignment

PASS

---

```text id="14qvmd"  
Gateway Assignment

PASS

```text id="ep0j8u" DNS Assignment

PASS

---

```
```text id="k1fm8j"
Search Suffix Assignment
```

PASS

Day 03 Conclusion

Day 03 successfully integrated enterprise DNS and DHCP services.

The final environment supports:

- Automatic client provisioning
- Centralized DNS management
- Internal hostname resolution
- Reverse DNS lookups
- Public internet DNS resolution
- Enterprise troubleshooting workflows

The infrastructure now resembles a small enterprise network where DHCP, DNS, routing, and client systems operate together as a unified service platform.

Status:

```
```text id="kptz0o" DAY 03 COMPLETE
```

```
```text id="qfd4eu"
DNS + DHCP INTEGRATION SUCCESSFUL
```